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Convergencia uniforme de funciones

Definición 1 (Convergencia uniforme). Sean $\forall n \in \mathbb{N} : f, f_n \colon X \longrightarrow Y$ con (Y, d) un espacio métrico, $(f_n)_{n\in\mathbb{N}}$ converge uniformemente a f en X,

$$\begin{aligned} f_n \xrightarrow[n\to\infty]{X} f &\iff \forall \varepsilon > 0 : \exists n_0 \in \mathbb{N} : \forall n \geqslant n_0 : \forall x \in X : d\left(f_n(x), f(x)\right) < \varepsilon \\ &\iff \lim_{n\to\infty} \sup_{x\in X} \Big\{ d\left(f_n(x), f(x)\right) \Big\} = 0. \end{aligned}$$

Referenciado en

- `teo-abel`
- `prop-convergencia-uniforme-imp-medida`
- `teo-cauchy-hadamard`
- `lem-convergencia-uniforme-imp-ctp`
- `convergencia-localmente-uniforme`
- `lem-convergencia-uniforme-esp-finito-imp-lp`

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